

Zomato Restaurant Data Analysis using Python

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UST Assignment 1

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1 Introduction

Exploratory Data Analysis (EDA) was performed on the Zomato Restaurant Dataset using Python libraries including Pandas, NumPy, Matplotlib, and Seaborn. The objective of this assignment was to understand the dataset, perform data cleaning, analyze restaurant information, visualize important trends, and derive meaningful business insights.

2 Dataset Overview

The dataset contains information about restaurants listed on Zomato. It includes restaurant details such as city, cuisine, pricing, ratings, customer votes, online delivery availability, and table booking services.

Dataset Summary

- Dataset loaded successfully using Pandas.
- Unnecessary index column (**Unnamed: 0**) was removed.
- Dataset shape and column information were explored.
- Data types were verified.
- Missing values and duplicate records were checked.
- The **Average Cost for two** column was converted to a numeric data type.

3 Data Cleaning

The following preprocessing steps were performed before analysis.

- Removed the unnecessary index column.
- Checked for missing values.
- Checked and removed duplicate records.
- Converted numerical columns to appropriate data types.
- Generated descriptive statistics using Pandas.

4 Data Analysis

The following analyses were performed.

- Number of unique cities.
- City with the highest number of restaurants.

- Most common cuisine.
- Average restaurant rating.
- Restaurant with the highest number of customer votes.
- Average cost for two people.
- Rating comparison based on Online Delivery.
- Rating comparison based on Table Booking.
- Top 10 cities with the highest number of restaurants.

5 Data Visualizations

5.1 Top 10 Cities by Number of Restaurants

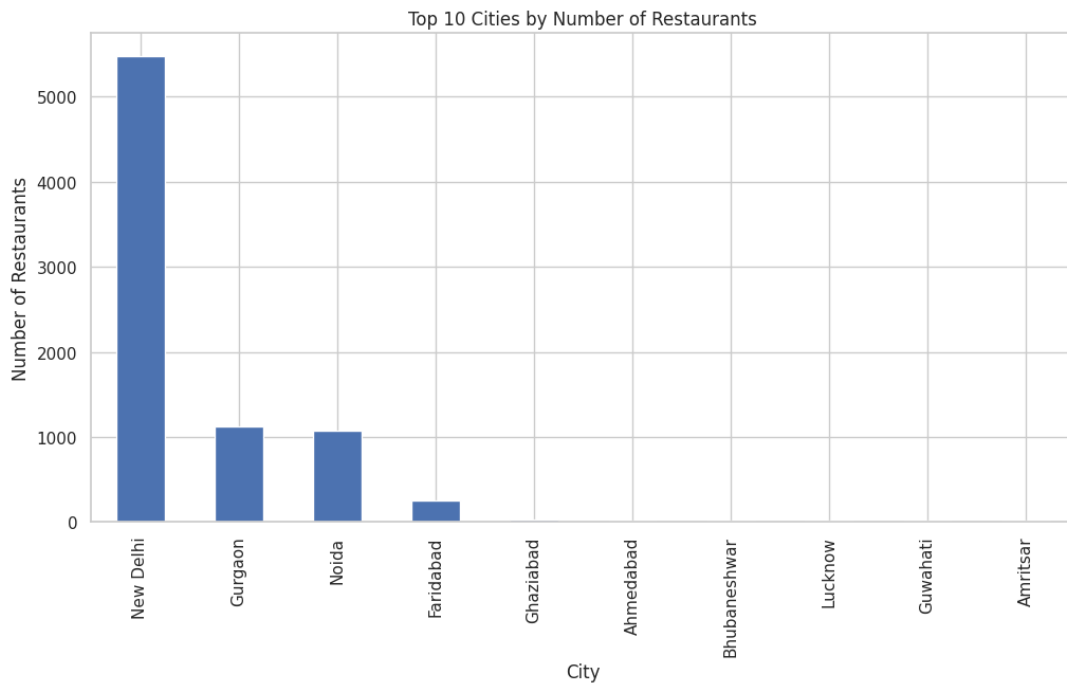


Figure 1: Top 10 Cities by Number of Restaurants

Interpretation

The dataset is highly concentrated in New Delhi, which has significantly more restaurants than any other city. Gurgaon and Noida are the next largest contributors, indicating that the Delhi-NCR region dominates the dataset.

5.2 Distribution of Aggregate Ratings

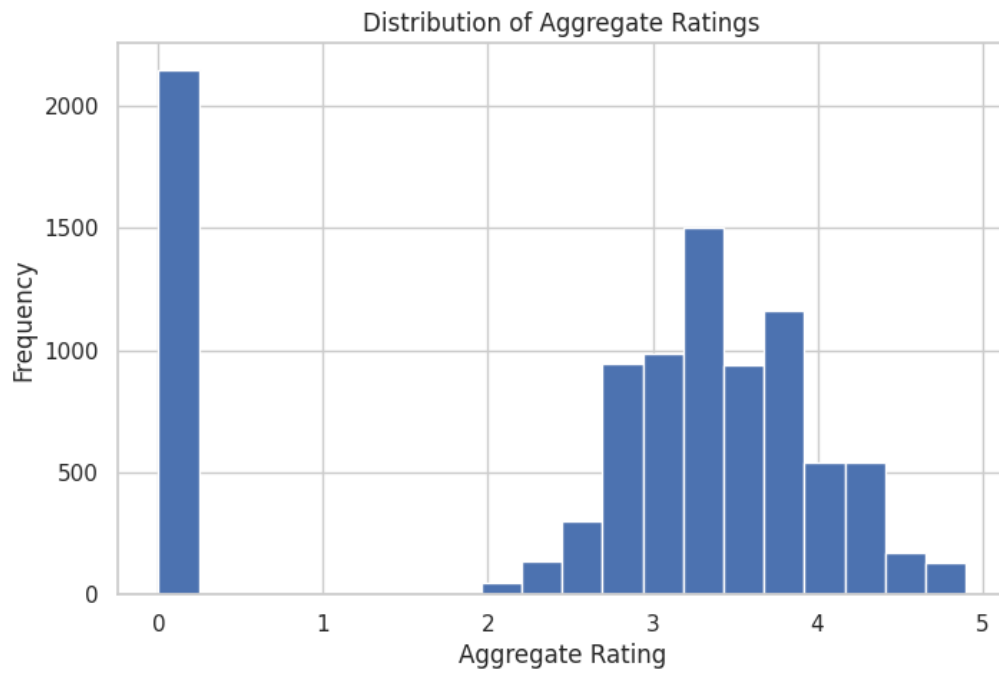


Figure 2: Distribution of Aggregate Ratings

Interpretation

Most restaurants have ratings between 3.0 and 4.0. A noticeable number of restaurants have a rating of 0, indicating either unrated restaurants or restaurants with insufficient customer reviews.

5.3 Restaurants Offering Online Delivery

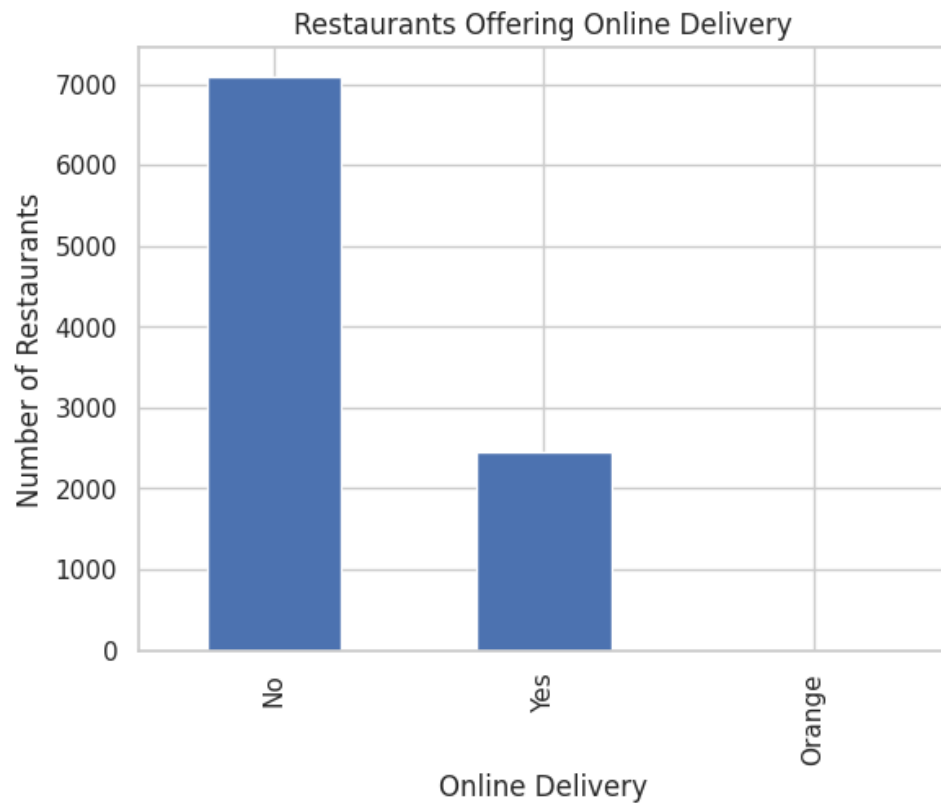


Figure 3: Restaurants Offering Online Delivery

Interpretation

The majority of restaurants do not provide online delivery. Restaurants offering online delivery represent a smaller portion of the dataset.

5.4 Top 10 Most Common Cuisines

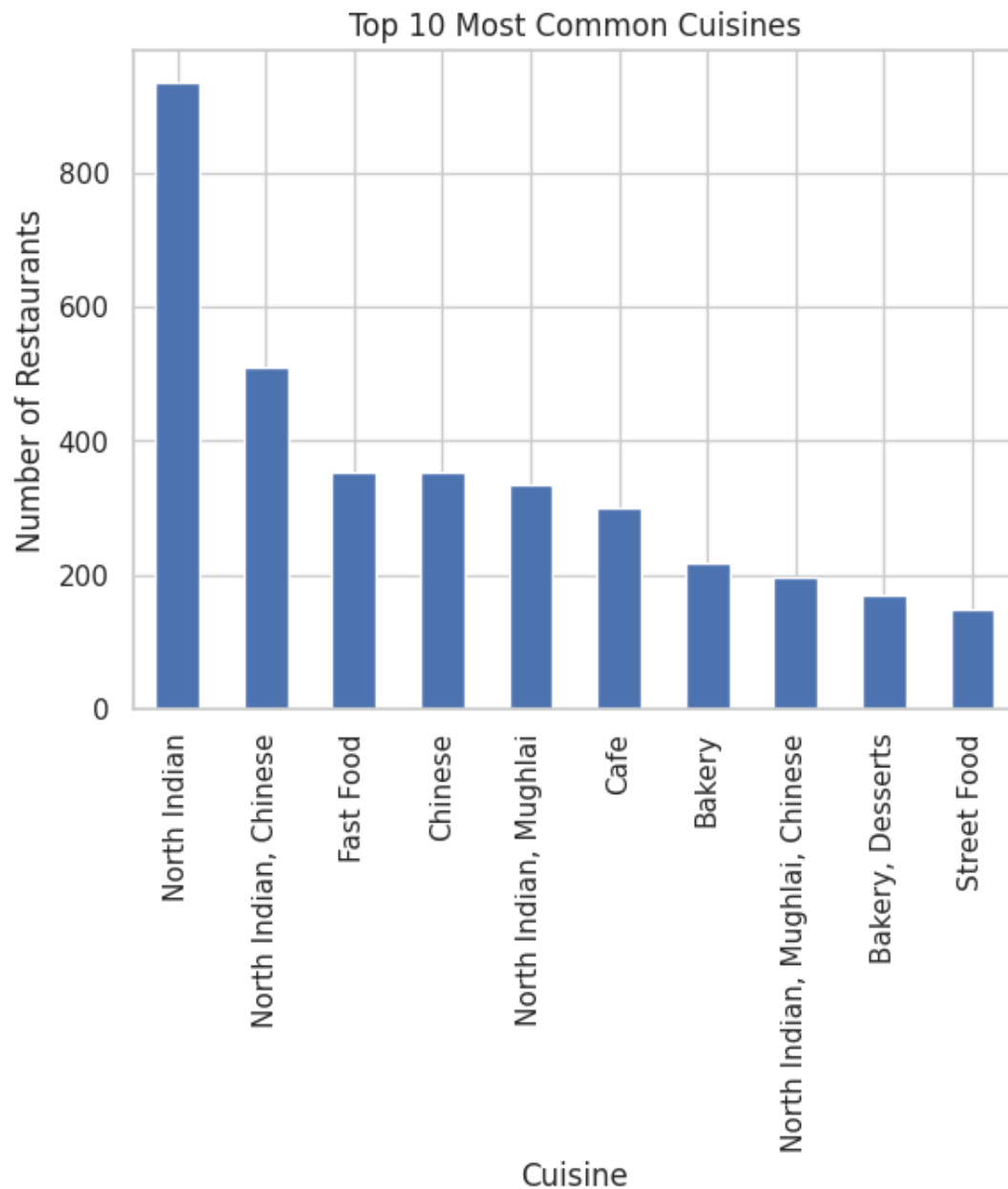


Figure 4: Top 10 Most Common Cuisines

Interpretation

North Indian cuisine is the most frequently available cuisine. Chinese, Fast Food, Mughlai, and Cafe-style restaurants are also highly represented.

5.5 Average Rating by Price Range



Figure 5: Average Rating by Price Range

Interpretation

Average ratings generally increase as the price range increases. Premium restaurants tend to receive better customer ratings compared to lower-priced restaurants.

6 Business Insights

The analysis provides several important business observations.

- New Delhi has the largest restaurant presence.
- North Indian cuisine is the most popular cuisine.
- Restaurants with online delivery can be compared to determine whether delivery influences customer satisfaction.
- Restaurants offering table booking generally receive higher ratings compared to those without table booking.
- Higher-priced restaurants generally receive better customer ratings.

7 Conclusion

This assignment successfully demonstrated the use of Python for Exploratory Data Analysis. The dataset was cleaned, analyzed, and visualized using Pandas, NumPy, Matplotlib, and

Seaborn. The generated visualizations highlighted important trends regarding restaurant distribution, customer ratings, cuisines, pricing, and service availability. The analysis also provided meaningful business insights that can help understand customer preferences and restaurant performance.